Training One Teaching Move into a Small Model

Geist Atlas · Module 16 · What Comes Next

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[**SCOPE-BOUND**] Supported within a narrow scope · **Family:** What Comes Next

Claim: Training produced partial traction but no solo arm passed its frozen gates. A live small-model-plus-frontier committee passed its scoped compliance endpoint only after checks covered every delivered-text path it owned; the hard-safety guardrail still failed.

The trained model nearly learned the missing warrant cue in its weights. In the final live run, the fully checked committee reached 0.477 (42/88) against 0.136 (22/162) for pooled controls, +0.341 with CI95 [0.223, 0.457], at no coverage cost and seam parity. The result is development-tier, one move, two worlds, one live tutor family, simulated learners, and no human-learning or deployment claim.

Derived view of `paper-full-2.0.md` v3.0.301 — §6.20, §6.21, §6.22. The paper is canonical; edit it there, not here.

6.20 Program-2: Training the Warrant Move into Small-Model Weights (Pre-Registered; Development-Tier)

§7.12 named one designed successor the closed adaptation programme could not answer from existing evidence: whether the insight–action gap is a context-versus-weights boundary — whether a model *trained* on the apparatus's own audit-labeled moments does what a model *told* about them would not. The experiment ran under a frozen pre-registration (`PROGRAM-2-PHASE2-PREREGISTRATION.md`; plan `PROGRAM-2-FINETUNE-PLAN.md` with the five HANDOFF revisions) as a two-arm design isolating the alignment layer: identical LoRA

*This sentence is the only one actually authored by Liam Magee in this paper.

data (865 audit-passing tutor turns harvested from the §6.18-addendum sealed traces, completion-only loss, seed 20260718) trained into Qwen3.5-9B-Instruct and its base sibling, graded on 58 held-out warrant-skip moments by the same deterministic machinery that produced the labels (offline grader fidelity against the 645 sealed verdicts: 99.83%/100%). Untuned floors were measured first and the pass bars frozen against them (floor + 0.15 with a paired-moment bootstrap, minimum absolute rate, and leak non-inferiority) before any training call. One floor observation stands on its own: a single model generation at like-for-like scale tripled the untuned floor (previous-generation 8B 0.121 → current 9B 0.362), with the older model failing exactly where the §6.18 frontier tutors failed — the warrant cue.

Verdicts (held-out greedy, n=58, same-serving-lineage pairs). The instruct arm reached **0.414 against its 0.310 floor** (+0.103; bootstrap CI95 [−0.017, +0.224]; bar 0.460) — fails the margin and uncertainty gates, passes the absolute-rate and safety gates: *partial parametric traction* under the frozen grammar. Inside that number, the trained skill's core nearly resolved: warrant-cue failures fell 19→6 on held-out moments (22→3 on dev) with conduct broadly intact — the first time in the programme that this frontier-resistant component moved by *any* intervention short of mechanical enforcement. The base arm stayed **flat at 0.103** with the cue trained just as strongly (43→9) but conduct collapsed (guard-pass 0.52→0.19) and leaks past tolerance — the thin-ego pathology the design predicted for the incumbent-free arm. **The iron-cage reading inverts:** the discipline trains into both variants' weights; what the incumbent alignment layer contributes is not resistance to the new norm but the stable surrounding conduct that lets it integrate. The cage is scaffold.

The coupling probe (pre-registered amendment). With the two failure profiles measured as complementary — the tuned mini has the cue and weak discipline, the frontier has discipline and no cue — a fail-closed protected-span composition was probed offline: the mini writes the demand sentence, a frontier model composes the turn around it verbatim, and any deterministic trip delivers the mini's own reply. Composed turns alone graded *below* the mini (0.293 — the frontier re-added extra questions even under explicit instruction; the checks, not the instruction, carry the design) *(attribution corrected by §6.22's decomposition of this same delivered file: the surplus questions were span-borne — the extraction joined all of the mini's question sentences — and the operative loss was extraction-dropped cue sentences; zero composer-added questions; the checks-carry-the-design conclusion stands)*, but the fail-closed system reached **0.448**, never below the mini by construction. The deployment-relevant contrast: the frontier tutors' own live compliance at these exact moments was **0.276** — the committee sizes at a +62% relative lift for one small-model call on trigger turns, with the live costs (the §6.18-addendum enforcement arm's ~0.13 coverage tax; seam visibility) explicitly unmeasured and reserved for a Phase-5 pilot. A structural note travels with all these rates: 14 of the 58 moments carry a due premise release and are non-compliant by construction under the frozen `released == 0` component, so the achievable ceiling is ≈0.76 and the tuned solo and committee figures correspond to ≈55% and ≈59% of achievable.

Status and caveats (per §5.12.6). Pre-registered gates with frozen bars, thresholds set before training; development tier — one world's moments, offline single-turn regeneration (no live-dialogue costs measured), n=58 held-out moments, q8/q4 quantized serving with per-arm lineage-matched floors, no human-learning claim, and no arm passed the frozen gates (no H-W claim). The two conditional KTO runs licensed here were subsequently spent (2026-07-21) and are behaviorally inert: KTO loss sat flat at 0.5 throughout — the unpaired audit labels supply almost no gradient the SFT policies do not already fit — and all 58 held-out greedy generations per arm are byte-identical to the SFT outputs, so every gate lands exactly where SFT left it (paired KTO-vs-SFT CI [0.000, 0.000] in both arms), the offline solo ceiling stands at the SFT number, and the Phase-2 licensed-run ledger (2 SFT + 2 KTO) is fully spent, with no further training licensed under that document (prereg results addendum). Four instrument faults occurred during evaluation and were each caught by a standing check before any wrong number was accepted — a partial file transfer, a converter/metadata inconsistency (loader refusal), silent no-op adapter merges from a model-class mismatch (exposed by 95% byte-identical outputs; merges now assert nonzero weight deltas), and ~5 points of GGUF build-lineage noise (removed by reflooring the instruct arm through its tuned model's own conversion pipeline). One further tuned-base behaviour is disclosed as part of the result, not an instrument fault: the tuned base arm emits empty generations on exactly the 14 held-out moments drawn from compiled-constraint dialogues (0/58 for its untuned floor) — a training-induced degenerate mode on one prompt family, consistent with the thin-ego collapse (empty replies grade non-compliant and are counted in the denominator). Provenance: prereg results addendum (§10) and `config/adaptive-tutor-evidence/program-2-phase4-results.manifest.json` (artifact SHA-256s); training on one Lambda H100 (~US$8), all grading local and deterministic; board card `program-2-context-vs-weights-finetune`.

6.21 Program-2 Live: the Committee's Offline Gain Vanishes over One Unchecked Path, Survives Once the Battery Covers Every Delivered Text, Transfers to a Second World Unchanged, then Closes the Delivery Gap Its Own Census Locates (Pre-Registered; Development-Tier)

§6.20 reserved the committee's live costs — coverage under enforcement scale, seam visibility — for a Phase-5 pilot. That pilot and its confirm run have now sealed under their own frozen pre-registrations, and together they read as a single controlled experiment about where a deterministic check must stand. The Phase-5 design (`PROGRAM-2-PHASE5-LIVE-PILOT-PREREGISTRATION.md` §§2–3; results §9): 24 live dialogues on world-005-marrick under the §6.18-addendum operational spec verbatim — two trap profiles × six repeats per arm, seed 20260718, one tutor family (sonnet-5; the codex-family committee out of scope). The *silent-control* arm is the frontier tutoring alone with the frozen detector logging every assignment it would have made. The *committee* arm is identical except that at detector-fired warrant moments the first draft is produced by the §6.20 chain live: the tuned instruct mini (the Phase-4 artifact on its frozen serving pin) writes a reply greedily under the frozen activation instruction, its question sentence is extracted by the probe-identical rule, the frontier composes its turn around that span verbatim, and a fail-closed pre-delivery battery — non-empty text, verbatim span containment, exactly one question in the composed turn (the last check added and frozen before launch) — decides; any battery failure delivers the mini's own reply, and the chosen draft then flows through delivery guards identical across arms. Runtime provenance (prereg §7, Amendment 1): the first launch, on the current main-derived stack, dead-ended three consecutive dialogues at turns 2–3 on guard hardening that postdates the detector freeze and every reference number, and aborted with zero dialogues sealed; execution moved to the claim-run runtime itself (the §6.18-addendum launch lineage `91b8a50e`) with the Phase-5 machinery cherry-picked on top, restoring the stack match with the offline references and the training corpus. 24/24 dialogues sealed with zero attrition, and the pre-declared density precondition passed (75 committee and 81 control warrant opportunities; prereg §9).

Phase 5: the offline gain does not survive live coupling, and the loss is located to a single unchecked path. The primary endpoint fails: pooled live warrant compliance 0.200 (15/75) committee vs 0.160 (13/81) control — +0.040, bootstrap CI95 [−0.054, +0.133] straddling zero (prereg §9; manifest `config/adaptive-tutor-evidence/program-2-phase5-live-pilot.manifest`). The offline composite reference (0.448 against the frontier's own 0.276 on the same archived moments, §6.20) does not survive live coupling at this n. The component decomposition, however, locates the mechanism precisely rather than diffusely. The trained warrant-cue form *does* transfer live — cue component 0.560 vs 0.395, +0.165 over control, the fine-tune's signature visible in delivered dialogue — and is then consumed by a question-discipline leak, 0.720 vs 0.938 (−0.218), under a compliance rule that is a conjunction. The leak sits exactly on the one delivered-text path the battery never inspected: fallback-delivered mini replies, which carried a second question often enough that 24 of 75 committee moments shipped as `fallback_multi_question` (prereg §9). Premise timing and guards are arm-symmetric. The two cost questions §6.20 reserved both come back clean at this n: coverage@16 is 0.611 vs 0.625 (−0.014, CI [−0.083, +0.056]) — no signal of the §6.18-addendum enforcement arm's ~0.13 tax at the point estimate, though the interval cannot exclude a small one — and the pre-registered seam battery lands at chance, 18/36 = 0.500 detection accuracy against the frozen 0.65 bar (realized n: 16 committee-turn windows, 20 all-frontier), with identical mean continuity (3.75 vs 3.75). Hard-safety pass rates are identical across arms (0.58 vs 0.58) and leak counts arm-symmetric (prereg §9).

Two exploratory analyses attach to the Phase-5 close-out, labeled as such and licensing nothing (prereg §9). First, the fallback-trim counterfactual: re-scoring the fallback turns with their text trimmed to the first question yields committee 0.347 vs control 0.160 — eleven moments recovered — naming the unchecked fallback path as the single highest-value fix. Second, the cue-lexicon sensitivity rescore: under a relaxed compliance rule whose cue list is a fifteen-word lexicon derived mechanically from the world specification (frozen mid-run), the *control* arm scores 0.469 against the committee's 0.320 — the frontier's natural questioning anchors claims to scene-named evidence at a rate the frozen six-word rule undercounts, so the committee's frozen-rule edge is specifically the trained six-word form: the audit's letter, not a general superiority in evidence-anchored questioning. Training on audit exhaust taught the audit's letter; the live loop makes the distance between that letter and the move's spirit visible. By explicit decision (2026-07-20) the frozen v1 rule remains the arc's only scoring instrument; the relaxed-rule numbers are reported descriptively and license no reading.

Phase 5b: extending the same battery over the fallback path flips the verdict, and moves only the component it aims at. The confirm run (`PROGRAM-2-PHASE5B-FALLBACK-BATTERY-PREREG` §2; results §8) changes exactly one thing: when composition fails the battery at a committee moment, the fallback is no longer the greedy mini reply shipped unchecked but resolves through the same fail-closed logic — ship the greedy reply if it passes a fallback battery (exactly one question and the frozen cue present); otherwise resample the mini up to twice at its frozen sampled-pin temperature (0.35) and ship the first passing sample; otherwise trim the greedy reply to a single cue-preserving question sentence. Composed-path semantics, span extraction, and the control arm are byte-unchanged. Twelve committee-v2 dialogues and six fresh silent controls sealed (18/18, zero attrition, seed 20260720), and a frozen stationarity check licensed pooling the fresh controls with Phase 5's twelve (control n = 18 dialogues). The primary endpoint passes: 0.386 (32/83) vs 0.150 (18/120) — +0.236, CI95 [0.128, 0.354] (prereg §8; manifest `config/adaptive-tutor-evidence/program-2-phase5b.manifest.json`). The action is surgical: `exactly_one_question` moves 0.720 → 0.976 — now above the frontier's own 0.938 — while the warrant cue (0.560 → 0.542), premise timing (0.587 → 0.590), and guards (0.960 → 0.988) stay put. The fallback ledger records how: 19 resample rescues, 8 cue-preserving trims, 4 greedy passes, 1 unchanged (prereg §8). The live rate matches §6.20's offline composite reference (0.448) to within the arms' interval, and sits at ≈65% of the structural ceiling its own moment population imposes — 34 of the 83 committee-v2 moments carry a due premise release and are non-compliant by construction under the frozen `released == 0` component, capping the achievable rate at 49/83 ≈ 0.59 (≈51% against §6.20's offline ceiling of ≈0.76, whose held-out moments carry due releases at roughly half the live density — 24% vs 41%). The guardrails repeat Phase 5's cost picture: coverage@16 0.611 vs 0.639, within the frozen −0.05 margin — the enforcement-scale tax remains absent under the stronger committee — and the seam re-check (required, since trimming alters delivered texture) lands at 0.600 (24/40) against the same 0.65 bar on the full 20+20 window battery, with continuity 3.45 vs 3.75: trimmed fallbacks are slightly more visible than composed turns, and under the bar (prereg §8).

One guardrail formally fails, with an anatomy that points away from the committee. Phase 5b's hard-safety rate is 0.42 vs the pooled control's 0.61, against a ≥ control − 0.10 bar: a formal FAIL, reported as the frozen rule requires (prereg §8). The anatomy: every committee-arm safety failure is a single leak at turn 9; the identical turn-9 signature appears in the control arms of both runs (4 of 12 Phase-5 controls, 2 of 6 fresh 5b controls), predating the committee; and 4 of the 5 leaky committee turns are frontier-authored. The reading recorded with the verdict is a world property — the turn-9 due premise's release staging trips the leak audit stochastically in every arm — amplified here by one to two dialogues of arm noise. A successor run wanting a clean safety guardrail should fix the turn-9 staging, not the committee. (Phase 5's safety guardrail, on the same world, passed with identical rates across arms.)

Read together under the pre-registered grammar, the pair licenses one sentence (5b prereg §5): the Phase-5 loss *was* the unchecked fallback path — with the battery closed over every delivered text, span-level form-ownership beats the frontier live, at no coverage cost and at seam parity (exploratory tier; one family, one world). The pair also gives §7.11's substitution law a constructive corollary in miniature: between the FAIL and the PASS no model changed, no prompt changed, and no training occurred — what moved was where the checker stood. The committee's residual failures concentrated on precisely the delivered-text surface its battery did not reach, and extending the check to that surface moved that component alone, flipping the verdict. The committee's live behaviour is governed by the coverage of its checks, not by the fluency of its members — the conclusion §6.18 reached destructively for model-owned form, reached here constructively for harness-owned form.

Phase 5c: the validated system moves to a world it has never seen, and the gain moves with it — carrying none of the training world's vocabulary. The pair above cannot say what the fine-tune wrote into the weights: the warrant move as a *form*, or a memorized surface entangled with world-005-marrick's costume. The cross-world probe (`PROGRAM-2-PHASE5C-CROSS-WORLD-TRANSFER-PREREGISTRATION.md` §§2–5; results §9) freezes that question and moves the Phase-5b-validated system *unchanged* — same artifact, same serving pin, fallback policy v2, and an explicit no-KTO clause (a concurrent session was training KTO artifacts; one change at a time) — to `world_027_gazette_recall`, selected under a frozen rule: maximum costume difference from Marrick (non-period frame, contemporary newsroom-noir diction, zero derived-lexicon overlap) subject to structural comparability with the training world (8 premises / 7 rules against Marrick's 9 / 5, the same secret-plus-decoy architecture), which protects the frozen density precondition — the farther-costumed floor worlds (2–5 rules, ~10-turn design) were rejected by name as density gambles. Ten committee-v2 and eight fresh silent-control dialogues (seed 20260721), with the pooling rule inverted: Phase-5/5b controls are Marrick dialogues, so every contrast runs against the new world's own controls only. Seventeen of eighteen sealed; one committee dialogue is the programme's first attrition (both same-seed attempts died on the known auto-learner budget overflow — deterministic for that dialogue shape), leaving nine committee dialogues; the density precondition passed at 61 committee warrant opportunities, 6.8 per dialogue against Marrick's 6.9. The primary endpoint passes: 0.508 (31/61) vs 0.306 (15/49) — +0.202, CI95 [0.072, 0.338] (prereg §9; manifest `config/adaptive-tutor-evidence/program-2-phase5c.manifest.json`) — above the committee's own home-world rate (0.386), with `exactly_one_question` at 1.000 (the extended battery airtight in the new costume; control 0.857), the warrant cue at 0.754 vs 0.653, and guards at 1.000 in both arms. Both arms' cue rates sit far above their Marrick analogues — newsroom English natively speaks the frozen six-word letter ("record" above all) — and the committee's edge holds on top of that lifted floor. The pre-registered costume-leak metric — Marrick's fifteen-word mechanically derived lexicon minus the new world's lexicon minus the frozen six, counted in the mini-authored text of every delivered committee unit — reads **zero across all 61 units** (0.0 per 1k words against a home-world saturation reference of 31.8/1k computed pre-freeze on the sealed 5b spans; the control frontier itself scores 4 occurrences, all the generic word "fair"): the trained move arrived with none of the training world's vocabulary, cleaner of Marrick's lexicon than the frontier's own English on the same world. The seam battery lands at 0.515 (17/33; 13 committee + 20 all-frontier windows under the realized-n clause) against the frozen   0.65 bar, and hard safety is 0.89 vs 0.88 — far above the Marrick runs, with no analogue of the turn-9 signature, confirming the leak anatomy above as a world property rather than a committee property. One guardrail formally fails: coverage@16 is 0.489 vs 0.550, −0.061 against the frozen −0.05 margin — a miss of 0.011 whose bootstrap interval [−0.197, +0.053] spans zero at n = 9 — reported as the rule requires and carried as a caveat on the reading rather than excused. Under the pre-registered grammar the pass row applies, coverage-caveated: the trained warrant move behaves as a form, not a costume, and the move-library concept — train a small model on one move once, validate it on each new world with deterministic audits, retrain only on failure — is live at exploratory tier (one family, one transfer world). The first validation of a new world cost no training at all.

Phase 5d: a census of the sealed traces locates a second unchecked path — the delivered text is not the text the committee approved — and closing it lifts the rate again, moving only the components it targets. The 5b/5c pair leaves one surface uninspected, and a post-hoc census of the sealed committee traces finds it: at 47 of 83 Phase-5b committee moments the finalized turn that reached the learner is *not* the committee-approved envelope (all 34 due-release turns, whose text the clue-staging surface authors, plus 13 clue-staged turns where that surface overwrites the composed turn after the battery has already passed it), so the committee's approved text ships on only 36 of 83, and on the overwritten turns the audited cue word is lost in a discarded sentence. The rider (`PROGRAM-2-PHASE5D-DELIVERY-INTEGRITY-PREREGISTRATION.md` §2; results §9) adds two deterministic fixes and changes nothing else: **spanCue.v1** resamples the mini up to twice at its frozen sampled pin (0.35) until the question sentence it extracts carries a cue word (the live analogue of §6.22's offline span-mode v2), and **deliveryGuard.v1** — on a clue-staged turn only, where the finalized text differs from the approved envelope and no premise releases — swaps the staged turn's final question for the protected span verbatim, never touching a premise-release turn or the clue body. Twelve committee-v3 dialogues and six fresh controls sealed on world-005-marrick (18/18, zero attrition, seed 20260722), the fresh controls pooled with all eighteen archived Marrick controls (Phase 5's twelve plus 5b's six) under the frozen stationarity check (fresh 0.095 vs archived 0.150, Δ 0.055; control n = 24). The primary endpoint passes: **0.477 (42/88) vs 0.136 (22/162) pooled control — +0.341, CI95 [0.223, 0.457]** (prereg §9; manifest `config/adaptive-tutor-evidence/program-2-phase5d.manifest.json`), up from 5b's 0.386 and +0.236, and 42 of the 55 achievable moments ≈ **76% of the live due-release ceiling** (33 of 88 moments due-release, ceiling 55/88 ≈ 0.625) against 5b's 65%. Both mechanism endpoints pass: the guard-eligible delivered-cue rate is **13/15 = 0.867** (bar 0.75; 15 span swaps applied, 33 premise-release turns skipped, 37 shipped-as-approved), and the span cue-rate is **74/85 = 0.871** (bar 0.85; 5b baseline 0.37; spanCue ledger 44 greedy, 30 resampled, 11 miss). The costs stay where 5b/5c put them: coverage@16 0.625 vs 0.632 (within the −0.05 margin, no tax), and the seam re-check — mandatory because the guard alters staged-turn texture — lands at **0.550 (22/40)   0.65**, continuity 3.40 vs 3.75, the swaps invisible to a sonnet-class reviewer at window scale. One guardrail formally fails, with 5b's exact anatomy: hard safety 0.50 vs 0.63, the failures concentrated at turn 9 (safety-breakdown artifact: 5 of the 7 committee leaks and both control leaks fall on turn 9, and the committee leaks split 4 committee-authored to 3 frontier-authored — more mixed than 5b's, but the exoneration rests on the turn-9 concentration and the control arm's identical signature, not on authorship), arm-symmetric and world-bound — the same Marrick release-schedule leak 5c found no analogue of (world_027 safety 0.89), not a rider regression. Under the pre-registered grammar the pass row applies: with span resampling and the delivery guard closing the two leaks the census located, the committee controls warrant form on every text path it owns, at no coverage cost and at seam parity. The gain is delivery completeness, not pedagogy — the frozen six-word instrument stands, and a genuinely evidence-anchored question that happens to use none of the six words still scores non-compliant by construction; what 5d closes is the distance between the form the committee approves and the form the learner receives, which §7.11's substitution reading predicts should be worth exactly its coverage and no more.

Phase 5f: a terminal fresh-world probe is technically runnable but does not instantiate the endpoint. After three Tideway apparatus pilots bounded duplicate-clue and closure failures, a separately authorized, source-certified cohort froze eighteen new dialogues on `world_031_tideway_makerspace` (ten committee, eight fresh controls; seed 20260728) and a live opportunity-density stop. The stop fired after fourteen dialogues sealed — eight committee and six controls, zero final attrition, one recovered private-planner-boundary retry — because every profile-arm cell still contained **zero eligible `warrant_skip` opportunities**. Once all five `proof_skipper|committee` rows had sealed at zero, that cell's minimum of five was mathematically unreachable. The registered endpoint is therefore **not estimable: insufficient opportunities**: committee and control compliance are both 0/0, their difference and interval are undefined, and no transfer or costume-leak inference is licensed. Coverage and hard safety are 1.000 in both arms; the four remaining jobs were not launched and no seam review was run. The result does not negate Phase 5c's cross-costume transfer finding: it narrows its empirical support by showing that a fresh world's world/profile/detector combination may fail to instantiate the point of action at all. Residual fallback and closure burden remains visible but outside the treatment endpoint (the longest sealed row used 22 turns and 19 safe fallbacks). Provenance: `PROGRAM-2-PHASE5F-FRESH-TRANSFER-PREREGISTRATION.md` §13; manifest `config/adaptive-tutor-evidence/program-2-phase5f.manifest.json`; analyzer `scripts/analyze-program2-live-pilot-5f.mjs`; archive `~/.machinespirits-data/program-2/phase5f-live` board card `program-2-phase5f-fresh-transfer-world`. Exploratory, development-tier; no human-learning claim, DB, rubric, abstract, or headline-N change; no sections renumbered.

Status and caveats (per §5.12.6). All four runs are pre-registered with endpoints, bars, seeds, and reading grammars frozen before launch, at exploratory tier (no H-W claim at stake); development-tier evidence throughout — two worlds' trigger moments (one training, one transfer), one tutor family (sonnet-5), nine to twelve committee dialogues per run, no human-learning claim, and the offline references were generated on the archived moment distribution under a weaker battery (stated as references, not bars). Phase 5d's two mechanism endpoints (M1 guard-eligible delivered cue ≥ 0.75; M2 span cue ≥ 0.85) and its $\approx$0.55 rate reference were registered before launch; both mechanisms cleared their bars, and the point rate landed at 0.477 — below the $\approx$0.55 reference not because the ceiling was lower (this run's achievable ceiling, $55/88 \approx 0.625$, is in fact higher than the 5b ceiling, $49/83 \approx 0.59$, on which the $\approx$0.55 figure was built as $\approx$0.9 of achievable) but because the committee converted 76% of achievable moments rather than the $\approx$90% that figure implicitly assumed. The 5c compliance instrument remains the frozen six-word letter, and the transfer world natively speaks it — the cross-world gain is measured on a floor that letter-friendly worlds lift for both arms; and "transfer" here means across costume within the same genre (same detector, same trigger family, same simulated learner profiles), not across task types or triggers. Instrument provenance: all three runs executed on the pinned claim-run runtime (`91b8a50e` lineage) per Phase-5 Amendment 1 (prereg §7) after the main-stack launch dead-ended with zero sealed dialogues — 5c on that worktree's `e9b01bdd` with the world parameterized and the 5/5b plans verified byte-identical under the change; Phase-5b Amendment 1 (prereg §6) mirrored the speaking surface's existing duplicate-line recovery onto prompt-model surfaces after endgame verdict-sentence repeats killed two early attempts at turns 19–26 — arm-independent and comparability-neutral (sealed dialogues by construction never hit the fatal, so no sealed content, including the pooled Phase-5 controls, is affected). Retries: six (Phase 5), two (5b), one recovered plus the one attrition (5c), all on the known auto-learner prompt-budget overflow, affecting both arms; 5c additionally ran a pre-registered single paid smoke dialogue on the new world (go criteria met: coherent spans in costume, zero serving errors) before launch. The conditional KTO runs (§6.20) were subsequently spent and came back byte-identical to SFT — behaviorally inert (§6.20 status note; Phase-2 prereg results addendum) — so the tuned-mini artifact used in all three runs here is also the programme's final trained artifact. Phase 5d executed on the same pinned lineage (`91b8a50e`) with machinery `27aae3b7`, after two pre-launch amendments both caught by the launch gates with zero sealed dialogues at fix time: a claude-CLI update that dropped the bare `sonnet-5` model alias (mapped to `claude-sonnet-5` at the spawn boundary only, every frozen string byte-identical) and the two new committee-v3 flags being read but not registered in the runner's argument whitelist (aborting the first launch attempt at job 1 before any model call; the smoke gained a process-level parse gate). Provenance: `PROGRAM-2-PHASE5-LIVE-PILOT-PREREGISTRATION.md` (results §9; Amendment 1 §7), `PROGRAM-2-PHASE5B-FALLBACK-BATTERY-PREREGISTRATION.md` (results §8; Amendment 1 §6), `PROGRAM-2-PHASE5C-CROSS-WORLD-TRANSFER-PREREGISTRATION.md` (results §9), and `PROGRAM-2-PHASE5D-DELIVERY-INTEGRITY-PREREGISTRATION.md` (results §9; Amendments 1–2); manifests with artifact SHA-256s `config/adaptive-tutor-evidence/program-2-phase5-live-pilot.man` `config/adaptive-tutor-evidence/program-2-phase5b.manifest.json`, `config/adaptive-tutor-evidence/p` and `config/adaptive-tutor-evidence/program-2-phase5d.manifest.json` (the last including the seam and safety-breakdown artifact hashes); analyzer `scripts/analyze-program2-live-pilot-5d.mj` and safety breakdown `scripts/program2-phase5d-safety-breakdown.mjs`; sealed traces archived under `~/.machinespirits-data/program-2/`; census and decision record `notes/program-2/2026-07-22-cue-attrition-observation.md`; results memo `notes/program-2/2026-07-20` board card `program-2-context-vs-weights-finetune`. No DB, rubric, abstract, or headline-N changes; no sections renumbered.

6.22 Program-2 Offline Follow-ups: the Composer Seat Is Family-Invariant, and the Protected Span Carries the Audited Outcome (Post-Hoc, Development-Tier)

Two same-day offline probes interrogate the committee from a side the live runs could not: does anything in the §6.20–§6.21 record depend on *which* frontier model sits in the composer seat, and what exactly does the composed turn lose against the mini's own reply? Both reuse the §6.20 coupling-probe apparatus unchanged — the same 58 archived held-out warrant moments, the same sealed tuned-mini greedy replies, the same frozen deterministic grader — with the composer family parameterized (`scripts/program2-coupling-probe.mjs --composer`; the default remains the Phase 4 pin, byte-preserved). Instrument fidelity was re-validated at zero cost before any paid call: the mini-solo regrade, the archived sonnet delivered file, and the fail-closed union reproduce §6.20's 0.414, 0.293, and 0.448-with-2-rescued byte-exactly against the Phase 4 manifest. Roughly 160 frontier calls in total — 54 composer calls per leg across the three paid legs (the v1 terra flip and the two v2 re-compositions; the sonnet v1 reference is a zero-call regrade of the archived Phase 4 file, and the four no-span moments never invoke the composer); no pre-registration — every number below is descriptive.

The family flip: swapping the composer changes nothing the audit can see. With the composer moved from sonnet-5 to codex `gpt-5.6-terra` (the family that runs the live seams but had never composed), the headline constructions are identical: delivered grade 0.293 (17/58) in both families, fail-closed union 0.448 (26/58) with 2 rescued in both — and the identity is moment-level, not aggregate coincidence: 56/58 moments receive the same verdict, one flipping each way. Containment differs marginally in terra's favor (1 span lost verbatim vs sonnet's 3; by source 53/1/4 vs 51/3/4), and composed-row component failures are near-identical (one-question 14 vs 13, cue 22 = 22, premise 11 vs 12, guards 11 vs 9).

The decomposition locates the composed-alone penalty in the harness, not the composer — and corrects a §6.20 attribution. Two mechanical facts explain the invariance. First, *neither composer added a single question*: every `exactly_one_question` failure among composed rows traces to the extracted span itself carrying two or three questions, because the v1 extraction joins *all* of the mini's question sentences into one span (histogram over the 54 span-bearing moments: 39 one-question, 14 two-question, 1 three-question; composer-added question failures: 0 in both families). Second, the composed-alone penalty is *extraction-dropped cue sentences*: 25 of 54 spans carry none of the frozen six cue words, and in 20 of those the mini's full reply held the cue — in a statement sentence the question-only extraction discards. Composed cue failures run 22 in both families against the mini's own 6, with each composer spontaneously restoring a cue in only 3 of the 25 cue-free cases; the mini keeps its own cue sentence, which is why the mini solo (0.414) out-graded composed-alone (0.293) in both families. §6.20's parenthetical "the frontier re-added extra questions even under explicit instruction" is therefore corrected in place: on the same archived delivered file the surplus questions were span-borne and the operative loss was the dropped cue; the companion conclusion — the checks, not the instruction, carry the design — stands, and a dated erratum on the Phase 2 pre-registration's results addendum (whose wording the clause inherited) records the same correction at the source.

Span-extraction v2: the named fix converts the full measured headroom, in both families. The decomposition names its own remedy, and it is the span-side analogue of the Phase 5b fallback trim: `--span-mode v2` (default v1 byte-preserved; regression check: recomputed v1 spans match all 54 archived rows exactly) selects *one* question sentence, preferring one that carries a cue word, and carries the mini's cue-bearing statement sentence into the protected span when no question holds one. A deterministic precheck before any call confirmed the design reaches its targets: all 54 spans single-question, cue coverage 29/54 → 49/54, exactly the 20 predicted statements carried. Re-composed under v2 (~55 calls per family), delivered compliance doubles — **0.586** (34/58) sonnet, **0.603** (35/58) terra — and the fail-closed composite rises from 0.448 to the same figures, with rescued moments (composed passes where the mini solo fails) going from 2 to 10 and 11: the composition step flips from a liability the fallback contained into the system's working margin. Against the offline held-out set's own achievable ceiling (44/58 ≈ 0.76, §6.20), the composite moves from ≈59% to 77–79% of achievable. The composed turns now out-grade the mini solo (0.414) outright — under v1 they never did. Residual failure mass sits on premise timing (11–12) and guards (10–11), surfaces span extraction does not touch; containment tightens further (sonnet 53/1, terra 54/0); and the family-invariance holds under the new rule, 55/58 same verdicts.

One real asymmetry surfaces once the span noise is gone. With every v2 span carrying exactly one question, any two-question composed turn means the composer added one: sonnet did, in 3 of 53 composed turns; terra in 0 of 54. The corrected §6.20 clause is thus bounded rather than resurrected — a small, sonnet-specific added-question tendency exists under the v2 instruction, was invisible under v1 where span-borne failures swamped it, and is caught by the same deterministic battery either way. Terra under v2 is the cleanest composition configuration measured in the programme: zero span losses, zero added questions.

What the pair adds to the record is one measurement and one repetition. The measurement: the committee's design premise that the frontier member is interchangeable — rented fluency around owned, checkable moves — is now observed rather than asserted, across two frontier families and two extraction rules, with one net moment of difference; the audited outcome is pinned to the protected span's content, which the mini and the harness own. The repetition: §6.21's design law runs a third time in miniature, this time on the harness's own plumbing — the highest-loss unchecked surface was the extraction step between the organs, and extending cue preservation over it converted the full measured headroom (13 cue-only composed failures per family) without touching any model, prompt, or weight. The live side of family dependence — a terra tutor's own compliance floor, trigger density under a terra tutor, learner response to terra texture — remains unmeasured by explicit decision (2026-07-21): the offline question the flip was run to answer, whether composition discipline is family-bound, is answered no.

Status and caveats (per §5.12.6). Post-hoc exploratory, development-tier, no pre-registration — descriptive numbers only, no verdict bars, no H-W claim. Offline single-turn regeneration at archived moments: n=58, one world's moments, one mini artifact, and the letter-not-spirit boundary applies unchanged (the frozen six-word cue rule; §6.21's world-lexicon caveat). These offline rates are not live projections — the live moment population's due-release density differs from the offline set's (see §6.21's ceiling correction) — and the v2 extraction exists only in the offline probe: the live committee machinery on the pinned worktree is untouched, and adopting v2 live is a separate, unlicensed decision. Provenance: `notes/program-2/2026-07-21-terra-composer-probe.md` (probe and same-day v2 addendum); manifest with artifact SHA-256s `config/adaptive-tutor-evidence/program-2-terra-probe.manifes` machinery `scripts/program2-coupling-probe.mjs` (`--composer`, `--span-mode`) and `scripts/program2-terra-probe-analyze.mjs`; artifacts under `~/.machinespirits-data/program-2/floor/`; board card `program-2-context-vs-weights-finetune`; HANDOFF H9. No DB, rubric, abstract, or headline-N changes; no sections renumbered.

Addendum (2026-07-26): the selector seat closes — the paid per-turn classifier carries no stall signal, and what discrimination the gate has comes from a

deterministic position flag the tutor already computes. §6.20–§6.22 grade the *writer* seat: once the tutor speaks at a point of action, `auditTutorStubPointOfActionCompliance` scores exactly-one-question, warrant cue, no-new-premise and guards-passed. The *selector* seat — one frontier call per learner turn emitting `evidence_use`, whose value is the sole input to `warrant_skip` — was never graded at all, so a turn could score a perfect 4/4 while intervening there was pointless. A zero-call grader closes the gap (`services/pointOfActionGateGrader.js`), scoring gate decisions against proof-DAG advancement from the archived per-turn state observation, window = state(N+1) − state(N). Identification is free from the existing arm design: `standing_book`, `silent_control` and `committee` compute and log the gate's assignment and then inject nothing at trigger time (verified: 705 fired turns, `interruption.kind` null on every one), so on those arms flagged-vs-passed-over is a property of the gate with no treatment in the way — and the grader derives the arm classification from the traces rather than a hardcoded list, so a changed injection path surfaces as a mismatch instead of quietly entering the baseline. **The selector's label carries nothing.** Across 183 dialogues, 4,372 scorable decisions and 2,718 treatment-free observational moments, the warrant-aligned advance rate is 14.9% (126/844) on fired turns against 15.4% (288/1,874) on passed-over turns: +0.4 points, SE 1.5, z 0.30, replicating as nothing on each arm independently (`standing_book` +3.0, `silent_control` +0.9, `committee` −0.9) across two archives and two scenario suites. The sharper form of the same fact: firing on *every* turn gives a stall precision of 84.8%, the live rule 85.1% — the paid gate is 0.3 points better than having no gate. On the tutor-gated components the sign inverts and separably so (`grounded` −6.9, z −3.50; `coverage` −8.5, z −4.76), which the labels explain: `omits_warrant` and `overleaps_evidence` both describe a learner asserting conclusions, and an asserting learner is *moving*. The instrument is a "learner is running ahead" detector read as a "learner is stuck" detector, and the two are mildly anti-correlated. Acting on the flagged moments did not help either (`side_coach` −2.1, `triggered_placebo` −3.6, `compiled_constraint` −8.9 — the last confounded by its own `suppress_new_premise` enforcement, identified by `side_coach`'s absence of the same collapse). The label-side half of the question, run the same day on 398 uniformly drawn archived records re-asked under the repaired rubric, agrees and adds a second reason: the v2 edit moved the gate's own fire rate 44.5% → 41.2%, 1.35 SE from zero, and the edit accounts for almost none of the movement — records whose stored label was something the edit *cannot* apply to churned at 39.4% anyway, and 41 records moved *into* `omits_warrant` under a definition that can only move records out. A target that reproduces itself on ~60% of re-asks is not trainable, so the archive is closed as a label source independently of what the label predicts. One apparent signal in the whole exercise fails to replicate and is worth recording as a method lesson — `overleaps_evidence` as a solo rule scores 91.4% on the step4 archive against an 85.2% always-fire base (a 6.2-point margin, the only number that would have justified a live run) and 83.8% against 84.7% on program-2, i.e. slightly *worse* than firing on everything; the step4 cell is 93 fired moments, and a single-archive read would have produced a plausible, citable, wrong recommendation to narrow the gate to one label.

Where the discrimination actually lives, and why no live A/B follows. The live gate is a conjunction: the model's vote AND a deterministic position layer (`near_closure`, `close_inquiry`) computed from the same turn's DAG state. Splitting the conjunction on the window-eligible moments — the `[3, 24]` trigger window is a harness constant every candidate inherits, so it belongs in the denominator (n = 2,339) — puts the discrimination entirely on the deterministic half: the position layer alone fires on 1,736 turns (74%) at 86.6% stall precision, the model vote alone on 1,162 (50%) at 80.9%, the live conjunction on 844 (36%) at 85.1%, and always-fire on all 2,339 at 82.4%. `near_closure` read as a whole-population state feature is +16.3 points (z 8.03), replicated per archive. So the paid call has no positive incremental value in any stratum, and the position flag beats fire-on-everything by +4.2 (SE 1.1, z 3.71). It does *not* beat the incumbent: position minus live rule is +1.6 (SE 1.5, z 1.06), not separable from zero. Three facts then close the question of whether to test the cheaper policy live, and it is worth being explicit that the first was expected to point the other way. (i) There is no cost saving. `evidence_use` rides inside one combined extraction call the tutor makes anyway for the DAG state and the register policy, and `near_closure` derives from that same call's `bestPathCoverage`; dropping the label from the gate saves zero calls and zero latency. (ii) There is no margin over the incumbent to detect. (iii) The fire rate moves 36% → 74%, which is a different intervention regime rather than a drop-in swap, and cannot be rate-matched without reintroducing a tuning knob the record does not license (a cooldown sweep oscillates 81.1–92.3% over 376 fired moments — fishing, not calibration). What survives is narrower than a cost argument and should be stated as such: the position flag is deterministic and judge-family-invariant, whereas `evidence_use` reaches only 78.6% cross-family agreement (weighted 0.583) even after the v2 repair and reproduces itself on ~60% of re-asks, so removing the label from the gate would remove the standing caveat that the intervention rate is partly a property of the chosen judge and the sampled draw rather than of learner conduct. And the flag's own reading is modest by construction — `grounded` moves −18.9 on the very rows `near_closure` selects, so it reads *position on the path*, not a stalled learner. "Do not interrupt someone four-fifths of the way through a derivation" is a sound operating rule, not a read of an interior. Against the arc's four ToM-redundancy nulls (§6.8.5–§6.8.6, §6.10, §6.13.6) this one runs with the polarity reversed: there, added machinery re-encoded what the *model* already inferred from context; here a *model call* re-encodes what a *deterministic state feature* already shows. Same boundary, approached from the other side — and the consequence for §6.20–§6.21's programme is that the local-model case rests wholly on the writer seat, whose +0.236 (Phase 5b) and +0.202 (Phase 5c) gaps are scored on turns where the tutor spoke and are untouched by anything here.

Status and caveats (per §5.12.6). Post-hoc exploratory, development-tier, no pre-registration and no verdict bars; zero API calls (everything is read off archived traces and re-runnable on any future archive). The channel is off-policy by construction: the outcomes are the ones that followed the gate that actually ran, so it ranks how well a rule *predicts* stalls in recorded dialogue, not what a rule would *cause* — a candidate beating 84.8% by a real margin would justify a live run, and nothing tested here does. Two measurement traps are recorded because both were live: a failing *draft* leak audit is the guard working (and is redundant with the guard outcome, so excluding on it double-counts and discarded ~390 clean turns in an earlier pass — only the delivered-text audit describes what the learner read, 79 turns across both archives), and `accounting.guards.leak` reports that the guard was *enabled*, not that anything leaked. `voiced_derived` at a 15.2% base rate is sparse but not degenerate, and the 2–3× denser tutor-gated components show the same null with the opposite sign, so the finding does not hinge on the target choice. Simulated learners, two worlds' scenario suites, one detector version (`step4-frozen-2026-07-14.v1`, unmutated — the grader is additive and reads the frozen gate's own recorded suppression keys); no human-learning claim. Provenance: `notes/program-2/2026-07-26-gate-grader-result.md` (this result), `notes/program-2/2026-07-26-relabel-sample-result.md` (the label-side half), `notes/program-2/2026-07-26-stall-corpus-audit.md`; machinery `services/pointOfActionGateGrader.js` and `scripts/grade-point-of-action-gate.js` (`npm run program2:gate-grade`, `--check` for the structural integrity gate now run automatically after every Program-2 launch pass); archives `~/.machinespirits-data/step4-claim-runs-2026-07` and `~/.machinespirits-data/program-2`; board card `point-of-action-gate-grader`. No DB, rubric, abstract, or headline-N changes; no sections renumbered.

Neighbouring modules: Module 14 — Advice, Enforcement, and the Cost of Control; Module 12 — More Layers Do Not Mean More Benefit; Module 9 — The Human-Learning Question; Module 4 — How We Checked Every Claim.